16.05.24

OPENCAMPUS.sh

Einführung in Data Science und maschinelles Lernen

VERSIONIERUNG MIT GIT (TEIL 2) UND AKTUELLE ENTWICKLUNGEN IM BEREICH ML

- Ergänzung der Teams
- Besprechung der Aufgaben
- Einführung in Git Teil 2
- Aktuelle Entwicklungen im Bereich KI bzw. maschinelles Lernen

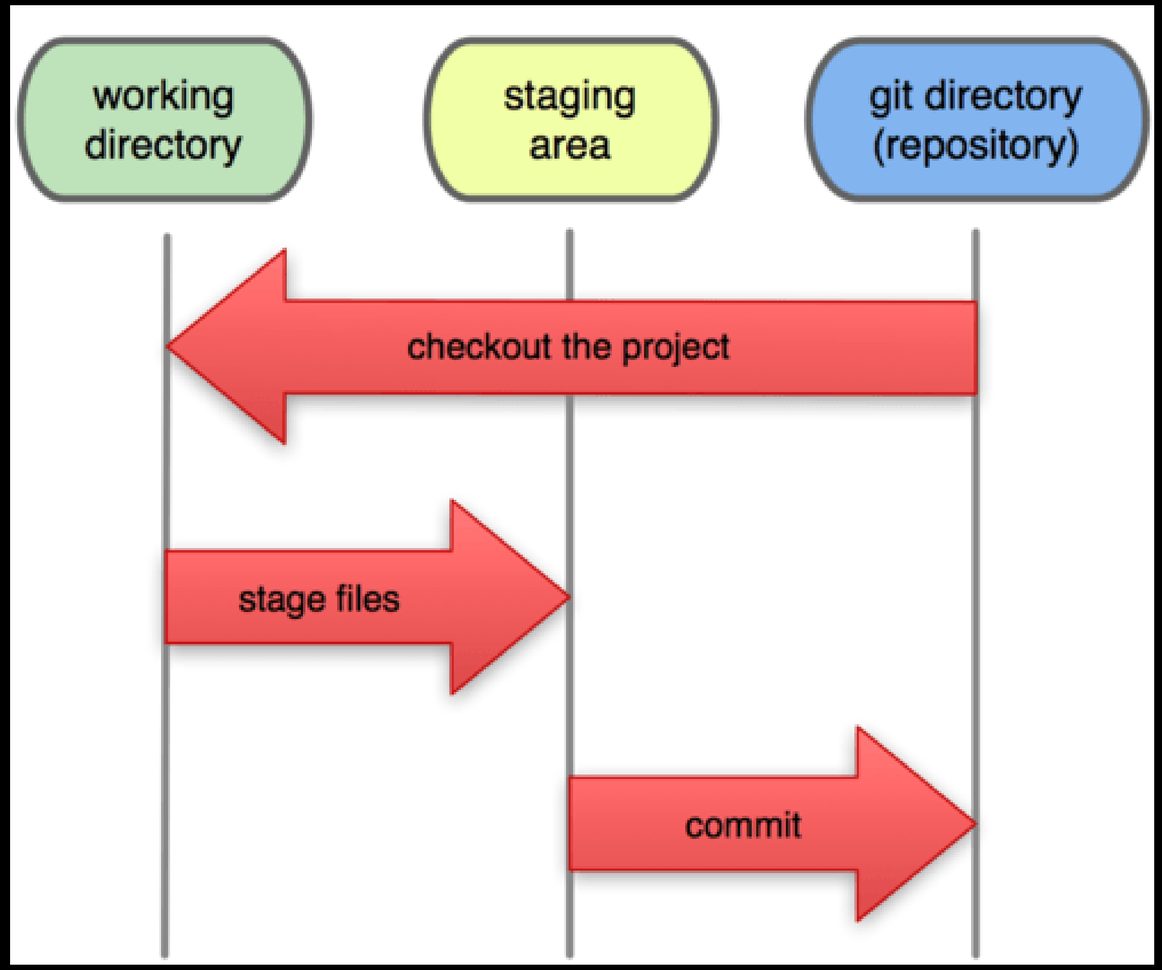
ERGÄNZUNG DER TEAMS

BREAKOUT

Besprechung der Aufgaben:

- **Eine Person im Team bestimmen, die ein Team-Repository anlegt.**
- **Einen GitHub-Codespace auf Basis Eures Team-Repositories anlegen.**
- **Die Dateien umsatzdaten_gekuerzt.csv, kiwo.csv und wetter.csv und in einem Pandas Dataframe zusammenführen.**
- **Euch im Team treffen und aufteilen, wer für welche Variablen deskriptive Statistiken und Visualisierungen erstellt.**
- **Deskriptive Statistiken und Visualisierungen zu den jeweiligen Variablen erstellen.**

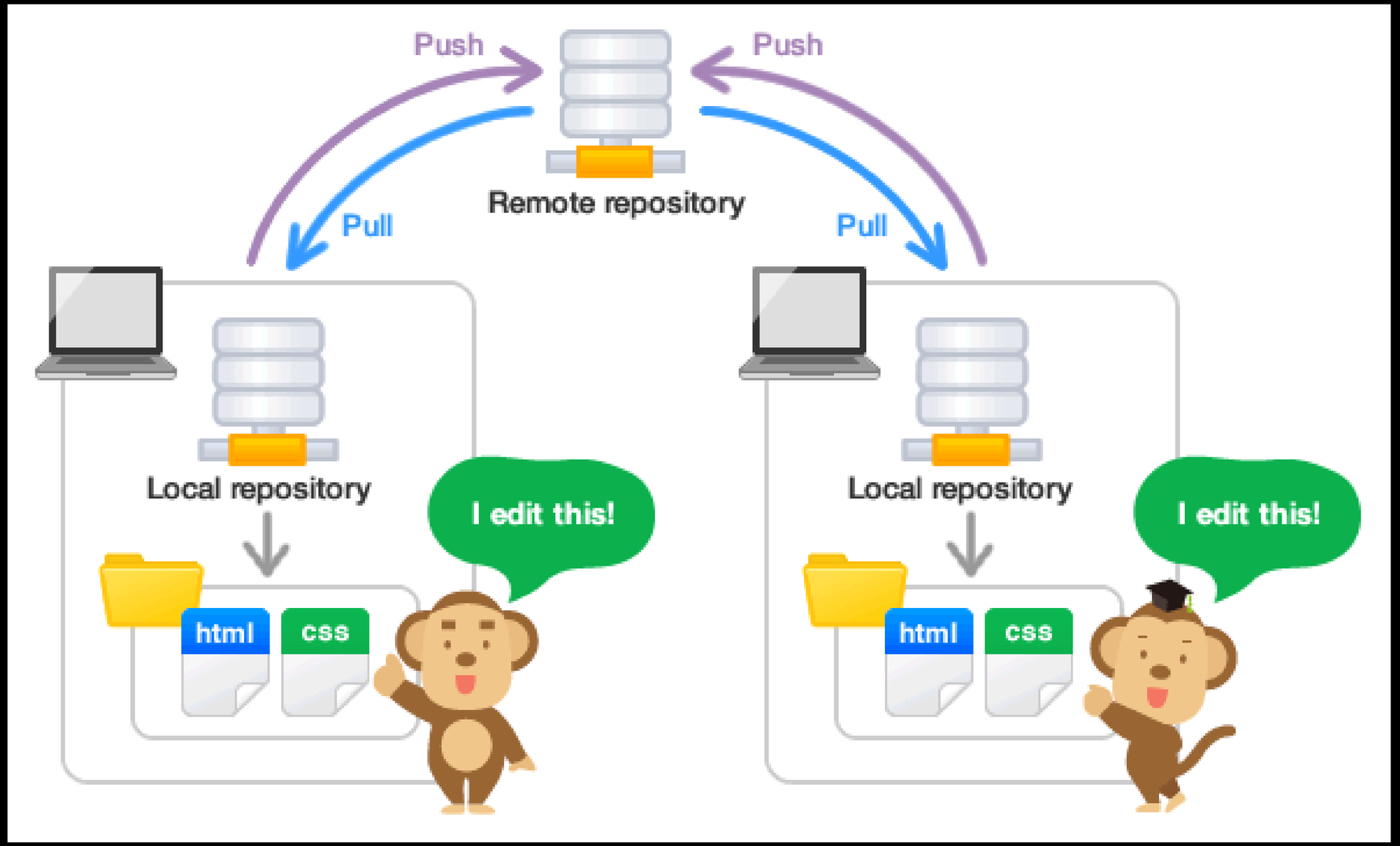
VERSIONIERUNG MIT GIT



Eine Datei kann drei mögliche Zustände haben:

- **modified („geändert“)**
- **staged („vorgemerkt“) und**
- **committed („versioniert“).**

LOKALE UND REMOTE REPOSITORIES



AUTHENTIFIZIERUNG

- Du bzw. die Anwendung auf Deinem Rechner benötigt Lese- und Schreibrechte auf dem Remote-Server.
- Dazu wird beim Lesen und Schreiben üblicherweise ein persönlicher Schlüssel verwendet, den nur Du bzw. Dein Rechner und der Remote-Server kennen.
- Über diesen Schlüssel kann der Remote-Server Dich und Deine Rechte auf dem Server identifizieren.

AUTHENTIFIZIERUNG BEI GITHUB

Variante 1, Du nutzt GitHub-Codespaces:

- **Die Authentifizierung erfolgt automatisch über Deinen GitHub-Account.**

Variante 2, Du nutzt ein lokales VS-Code:

- **Nach Installation der VS Code-Erweiterung „GitHub Pull Requests and Issues“ wirst Du automatisch aufgefordert, Dich bei GitHub einzuloggen. Dabei wird im Hintergrund ein privater Schlüssel für Dich bei GitHub erzeugt und bei Dir lokal abgelegt.**

DATENAUFBEREITUNG FÜR DAS MASCHINELLE LERNEN

- **Data Collection**
  Sammeln der Rohdaten aus verschiedenen Quellen wie Datenbanken, Dateien, APIs oder Web-Scraping.
- **Cleaning**
  Umgang mit fehlenden Daten, Entfernen von Duplikaten und das Korrigieren von Fehlern.
- **Data Exploration / Analysis**
  Verstehen der Verteilungen von Variablen, Erkennen von Ausreißern und Zusammenhängen., etwa durch statistische Kennzahlen, Korrelationsmatrizen oder Visualisierungswerkzeuge.
- **Feature-Engineering**
  Umwandlung von Rohdaten in Merkmale, die das zugrundeliegende Problem für die Modelle besser darstellen (etwa durch Erstellen neuer Variablen aus vorhandenen Daten), Kodieren von kategorialen Variablen und Behandlung von fehlenden Werten.
- **Data Splitting**
  Aufteilung der Daten in Trainings-, Validierungs- und Testdatensätze, um sicherzustellen, dass das Modell gut auf neue, unbekannte Daten generalisieren kann.

DATENAUFBEREITUNG

Bei jeder Modellierung müssen die Daten folgende Eigenschaften haben:

1. **Es darf keine fehlenden Werte geben.**
2. **Alle Werte müssen Zahlen sein.**
3. **Kategoriale Variablen sind one-hot-encoded (oder dummy-kodiert).**

ONE-HOT ENCODING

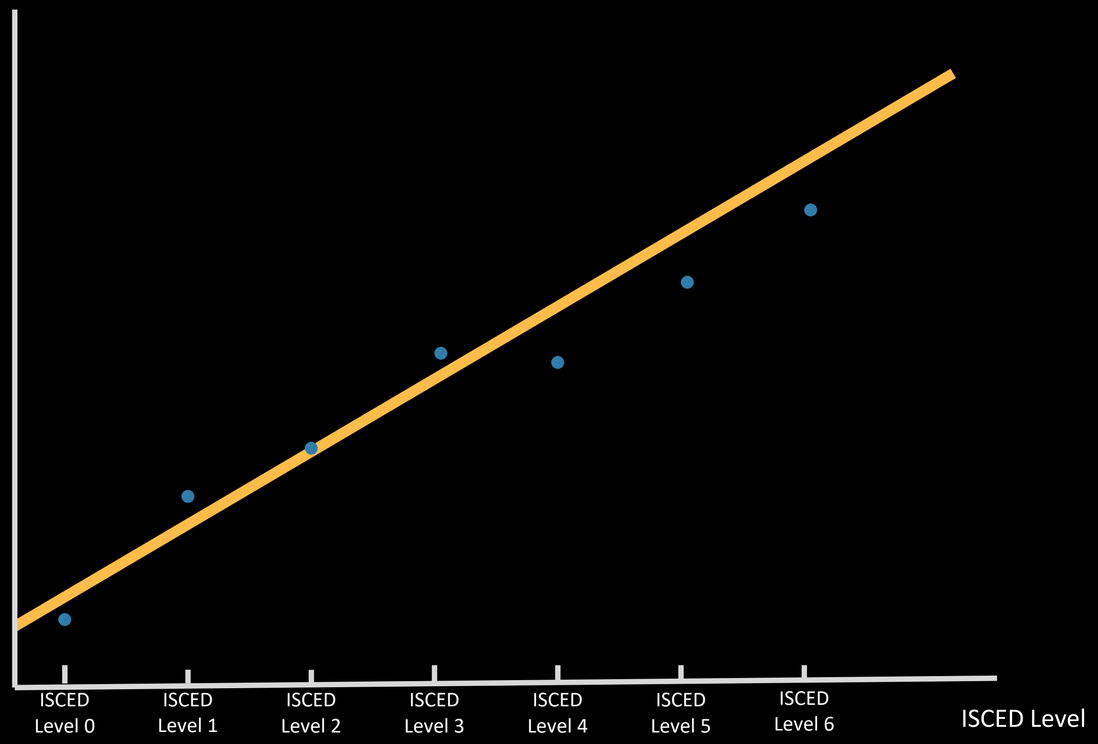
id	color
1	red
2	blue
3	green
4	blue

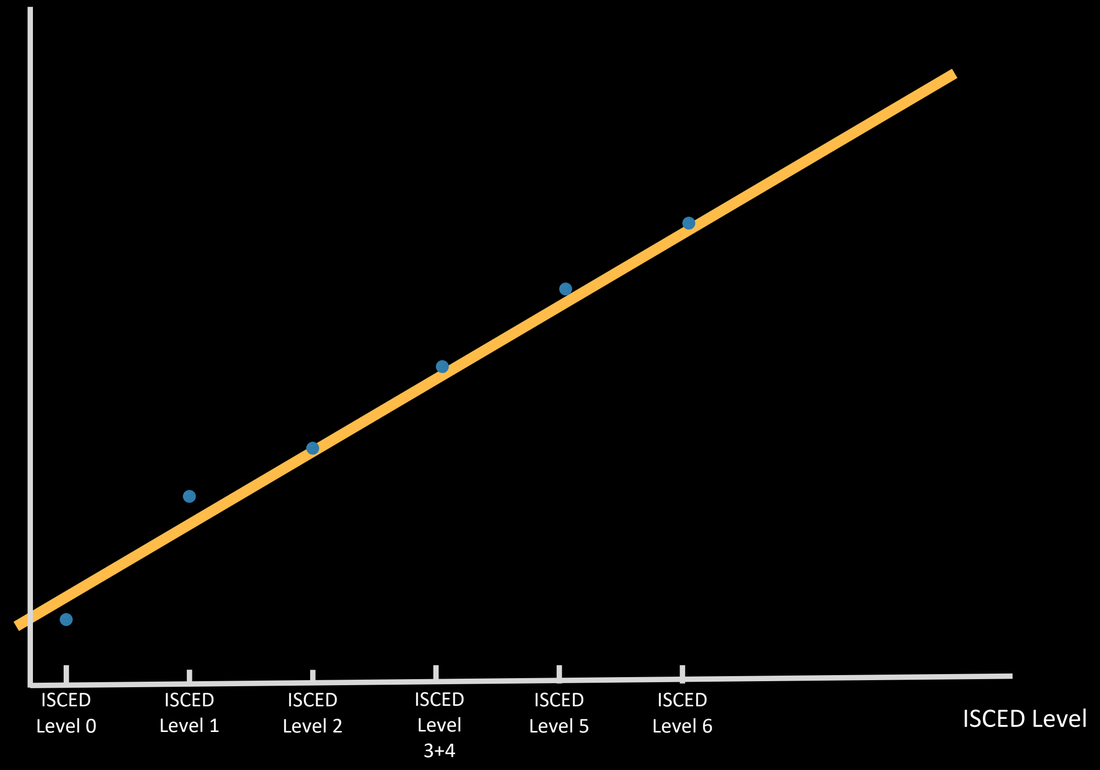


id	color_red	color_blue	color_green
1	1	0	0
2	0	1	0
3	0	0	1
4	0	1	0

VERBESSERUNG DER INTERPRETIERBARKEIT VON DATEN DURCH ZUSAMMENFASSUNGEN

VERBESSERUNG ORDINALER DATEN („BINNING“)





VERBESSERUNG NOMINALER DATEN

- Klassengrößen unter 50 haben üblicherweise sehr instabile Mittelwerte
- Sinnvolle Zusammenfassung von Klassen zu einer Mindestgröße von ca. 50
- Ggf. Nutzung einer „Reste-Kategorie“

BREAKOUT

- Welchen Zusammenhang hat die Tagestemperatur mit der Umsatzhöhe?

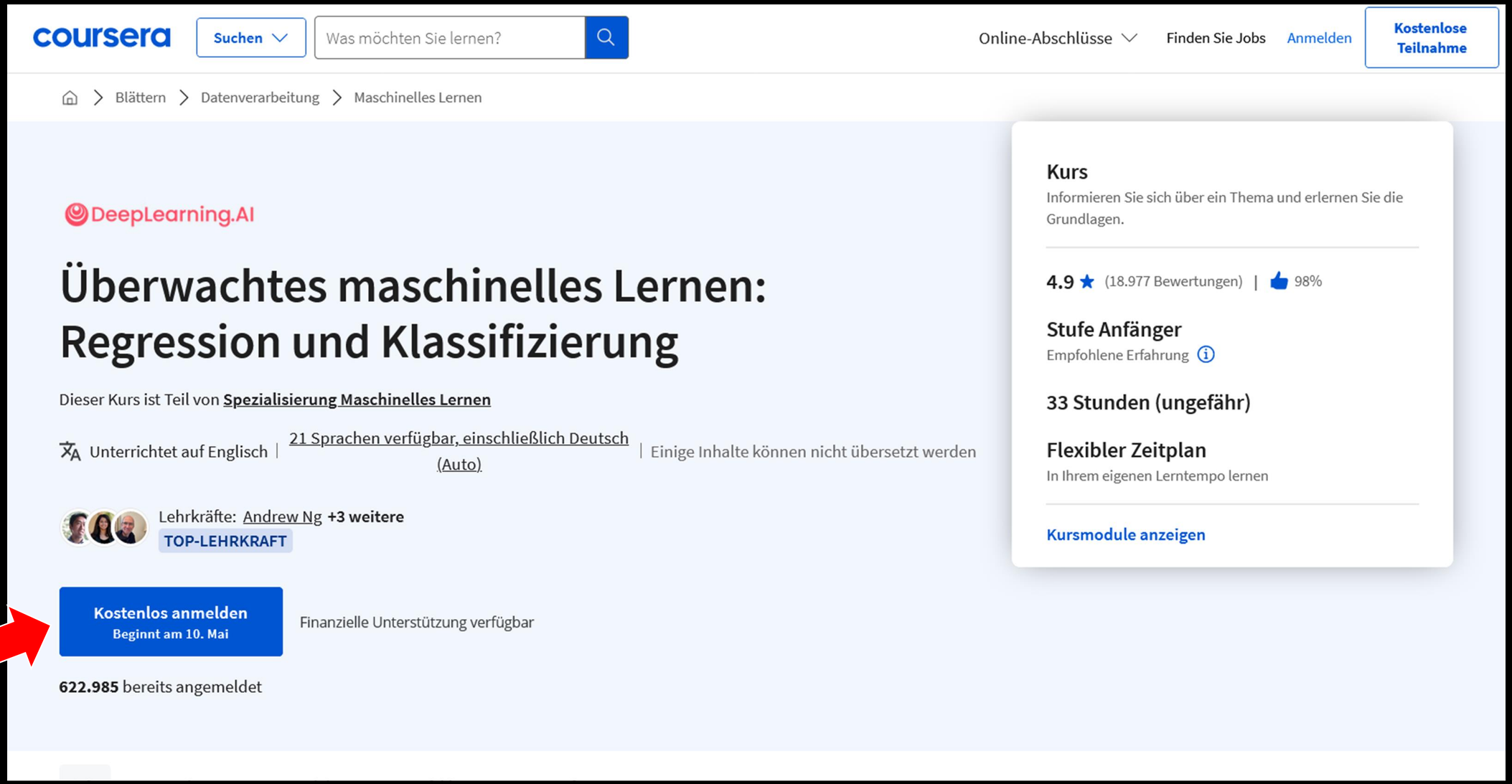
- Welche Möglichkeiten gibt es die Tagestemperatur als Variable im Sinne eines „Feature Engineering“ aufzubereiten?

LERNMATERIAL

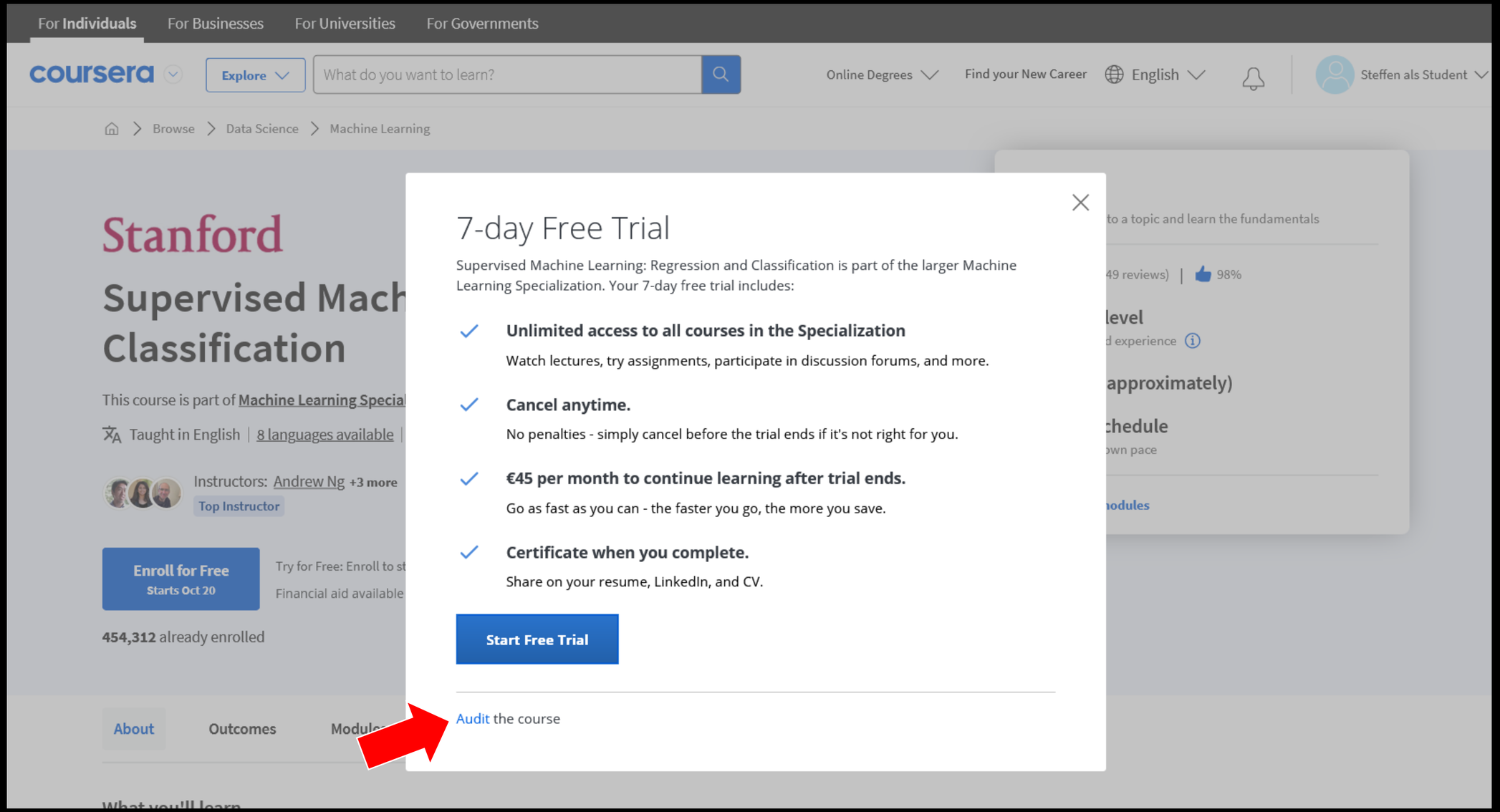
- **Absolviert das Kapitel „Creating Features“ dieses Kurses bei datacamp.**

- **Absolviert die erste Woche des Kurses Supervised Machine Learning: Regression and Classification (ca. 2,5 Stunden) von Andrew Ng auf Coursera.**

FÜR DEN KURS ANMELDEN



AUDIT MODUS AUSWÄHLEN



Hide menu

Overview of Machine Learning

Supervised vs. Unsupervised Machine Learning

Video: What is machine learning?
5 min

Video: Supervised learning part 1
6 min

Video: Supervised learning part 2
7 min

Video: Unsupervised learning part 1
8 min

Video: Unsupervised learning part 2
3 min

Video: Jupyter Notebooks
4 min

Lab: Python and Jupyter Notebooks
1h

Practice Quiz: Supervised vs unsupervised learning

Regression Model

Practice Quiz:



What is machine learning?

Save note

Transcript | Notes | Downloads

Transcript language: English

0:00 What is machine learning? In this video, you'll learn the definition of what it is and also get a sense of when you might want to apply it. Let's take a look together. Here's a definition of what is machine learning that is attributed to Arthur Samuel. He defined machine learning as the field of study that gives computers the ability to learn without being explicitly programmed. Samuel's claim to fame was that back in the 1950s, he wrote a checkers playing program. The amazing thing about this program was that Arthur Samuel himself wasn't a very good checkers player. What he did was he had programmed the computer to play maybe tens of thousands of games

Regression Model

Video: Linear regression model part 1
10 min

Video: Linear regression model part 2
6 min

Lab: Optional lab: Model representation
1h

Video: Cost function formula
9 min

Video: Cost function intuition
15 min

Video: Visualizing the cost function
8 min

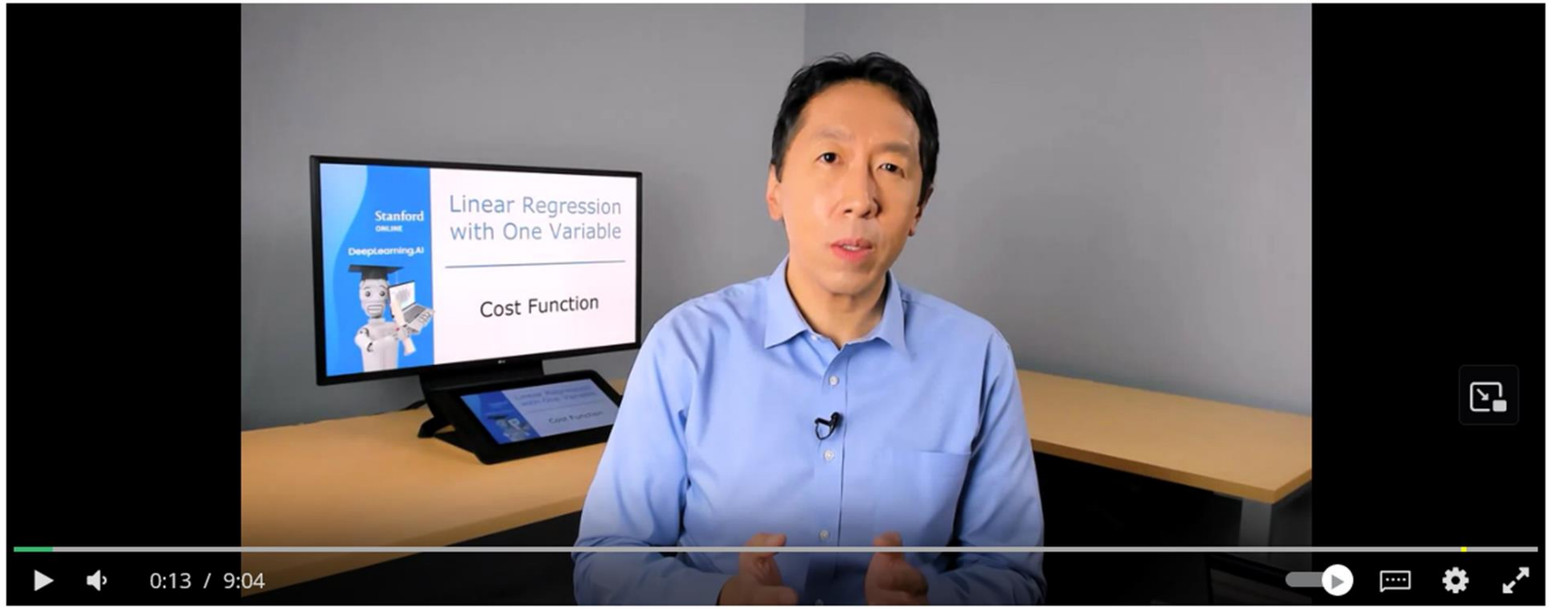
Video: Visualization examples
6 min

Lab: Optional lab: Cost function
1h

Practice Quiz: Regression Model

Train the model with gradient descent

Practice quiz: Train the model with gradient descent



Cost function formula

Save note

Transcript | Notes | Downloads

Transcript language: English

0:00 In order to implement linear regression the first key step is first to define something called a cost function. This is something we'll build in this video, and the cost function will tell us how well the model is doing so that we can try to get it to do better. Let's look at what this means. Recall that you have a training set that contains input features x and output targets y. The model you're going to use to fit this training set is this linear function f_w, b of x equals to w times x plus b. To introduce a little bit more terminology the w and b are called the parameters of the model. In machine learning parameters of the model are the variables you can adjust during training in

Unsupervised Machine Learning

Practice Quiz: Supervised vs unsupervised learning

Regression Model

Practice Quiz: Regression Model

Train the model with gradient descent

Video: Gradient descent
8 min

Video: Implementing gradient descent
9 min

Video: Gradient descent intuition
7 min

Video: Learning rate
9 min

Video: Gradient descent for linear regression
6 min

Video: Running gradient descent
5 min

Lab: Optional lab: Gradient descent
1h

Practice quiz: Train the model with gradient descent



Gradient descent

Save note

Transcript | Notes | Downloads

Transcript language: English

0:00 Welcome back. In the last video, we saw visualizations of the cost function j and how you can try different choices of the parameters w and b and see what cost value they get you. It would be nice if we had a more systematic way to find the values of w and b, that results in the smallest possible cost, j of w, b. It turns out there's an algorithm called gradient descent that you can use to do that. Gradient descent is used all over the place in machine learning, not just for linear regression, but for training for example some of the most advanced neural network models, also called deep learning models. Deep learning models are something you learned about in the

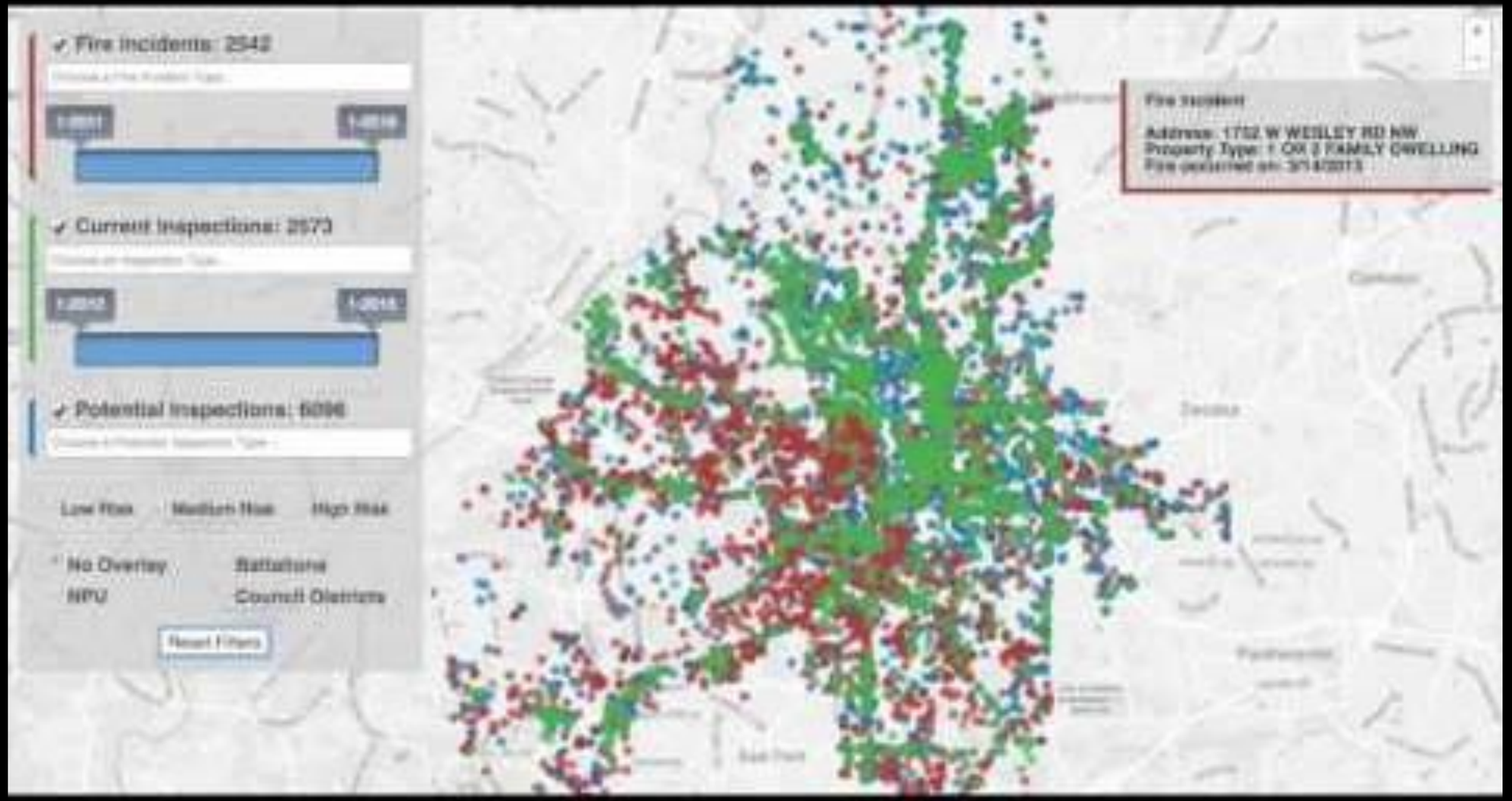
AUFGABEN

- Trefft Euch im Team, um mögliche zusätzlich zu erstellende Variablen zur Umsatzvorhersage zu diskutieren (auch, ob es Daten gibt, die ihr zusätzlich verwenden könnt).

- Ergänzt das Verzeichnis "Data Import and Preparation" Eures Team Repositories um:
  - zusätzliche heruntergeladene oder selbst erstellte Daten (z.B. Listen mit Feiertagen)
  - Code um alle Daten in einem Datensatz zusammenzuführen
  - Code, um neuen Variablen zu erstellen bzw. bestehende Variablen für die Vorhersage aufzubereiten

AKTUELLE ANWENDUNGEN IM KI- BEREICH

RISIKOPROGNOSEN



SCHACH-/ SPIELCOMPUTER



AlphaGo – The Movie:
https://www.youtube.com/watch?v=WXuK6gekU1Y

ROBOTERSTEUERUNG



BILD- UND VIDEOGENERIERUNG

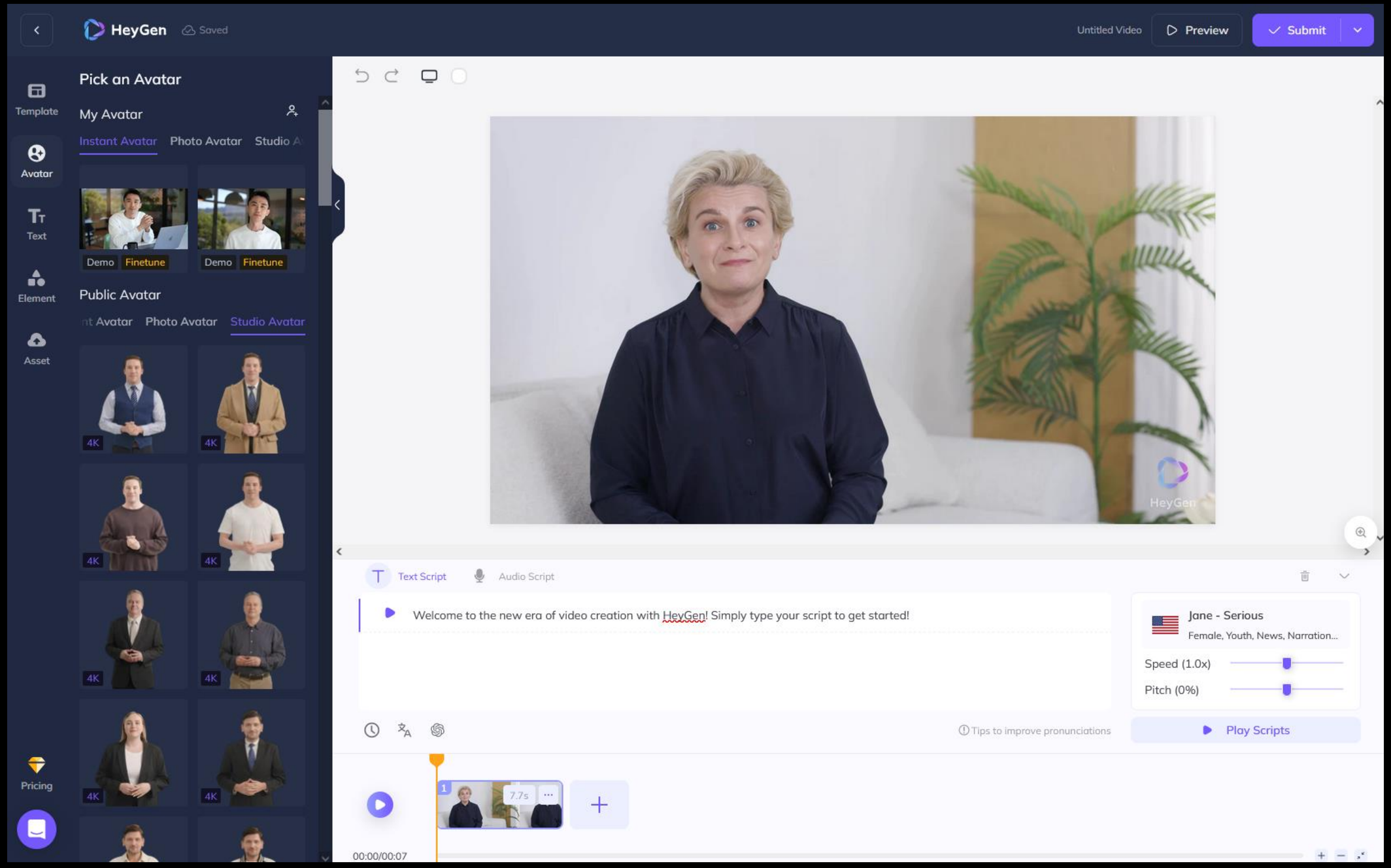




Advay Pal [@advaypal]. (2023, April 26). what have we done? Https://t.co/DVC0zL2Xgh [Tweet]. Retrieved October 9, 2023, from Twitter website: https://twitter.com/advaypal/status/1651355572980310016



HeyGen—AI Spokesperson Video Creator. (2023). Retrieved October 9, 2023, from https://app.heygen.com/home


HeyGen
Saved
Untitled Video
Preview
Submit
Template
Avatar
Text
Element
Asset
Pricing
Pick an Avatar
My Avatar
Instant Avatar
Photo Avatar
Studio A
Demo
Finetune
Demo
Finetune
Public Avatar
nt Avatar
Photo Avatar
Studio Avatar
4K
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4K
HeyGen
Text Script
Audio Script
Welcome to the new era of video creation with HeyGen! Simply type your script to get started!
Jane - Serious
Female, Youth, News, Narration...
Speed (1.0x)
Pitch (0%)
Tips to improve pronunciations
Play Scripts
1
7.7s
00:00/00:07

NATURAL LANGUAGE PROCESSING

GPT-3

Explain Kants Categorial Imperative

Share Try again

The Categorial Imperative is an ethical theory proposed by the German philosopher Immanuel Kant. This was one of two main moral theories in his philosophy (the other being Deontology). His 'categorical imperative' proposes that an action can be considered moral if it could potentially become a universal law.

The Categorial Imperative is essentially the idea that the ends don't justify the means. Only actions which are consistent with a universal law (that can be applied consistently to all human beings) can be considered moral.

The Categorial Imperative also suggests that the motivations behind an action are irrelevant. For example, if I was to kill my enemy in cold blood, but did it out of compassion and love for humanity, this would not make it moral - as the act itself is contradicting a universal law.

However, it is important to note that the Categorial Imperative also says that an action can be considered moral if there are no other means of achieving a goal. For example, if I was in need of money for my own survival and I needed to steal it from someone else - without causing any harm.

Overall, I would say Kant's Categorial Imperative is a moral theory that puts great emphasis on the principle of 'doing unto others as you would have them do to you'.

It is also worth noting that Kant's Categorial Imperative disagrees with the Utilitarian theory of ethics, as it puts more emphasis on an action being consistent with a universal law rather than providing the greatest good to the greatest number.

Schreiner, M. (2020, August 31). GPT-3: Diese fünf Apps zeigen das KI-Potenzial [Blog]. Retrieved May 4, 2021, from MIXED website: https://mixed.de/gpt-3-diese-fuenf-apps-zeigen-das-ki-potenzial/

CHATBOTS ALS AGENTEN

ChatGPT Plugins

How can I help you today?

Geben Sie mir Ideen
about how to plan my New Years resolutions

Zeige mir einen Code-Ausschnitt
eines Sticky-Headers einer Website

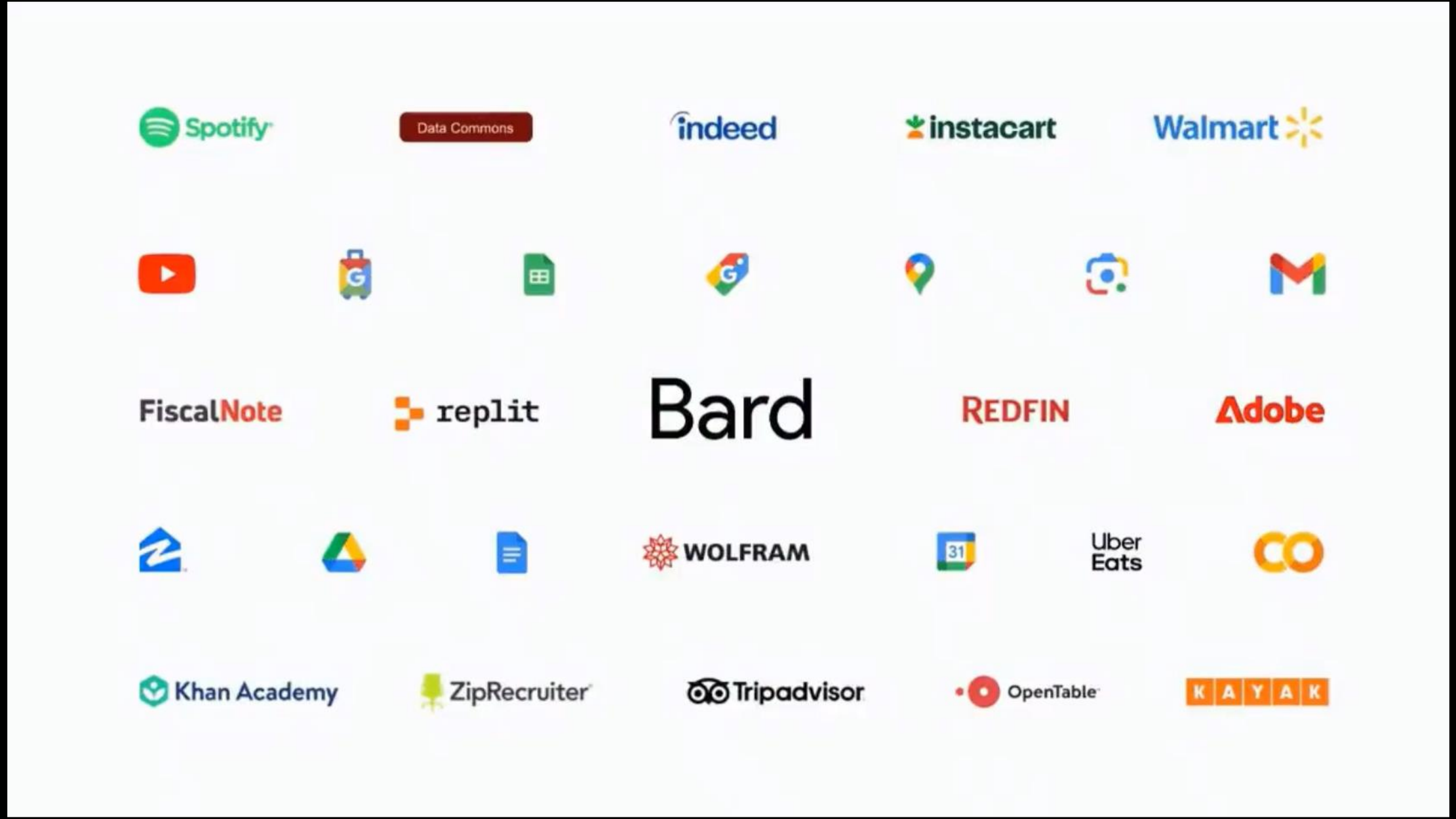
Erfinde eine Geschichte
über Sharky einen Zahnputzenden Hai-Superh...

Konzepte entwickeln
Für ein Retro-Arkaden-Spiel

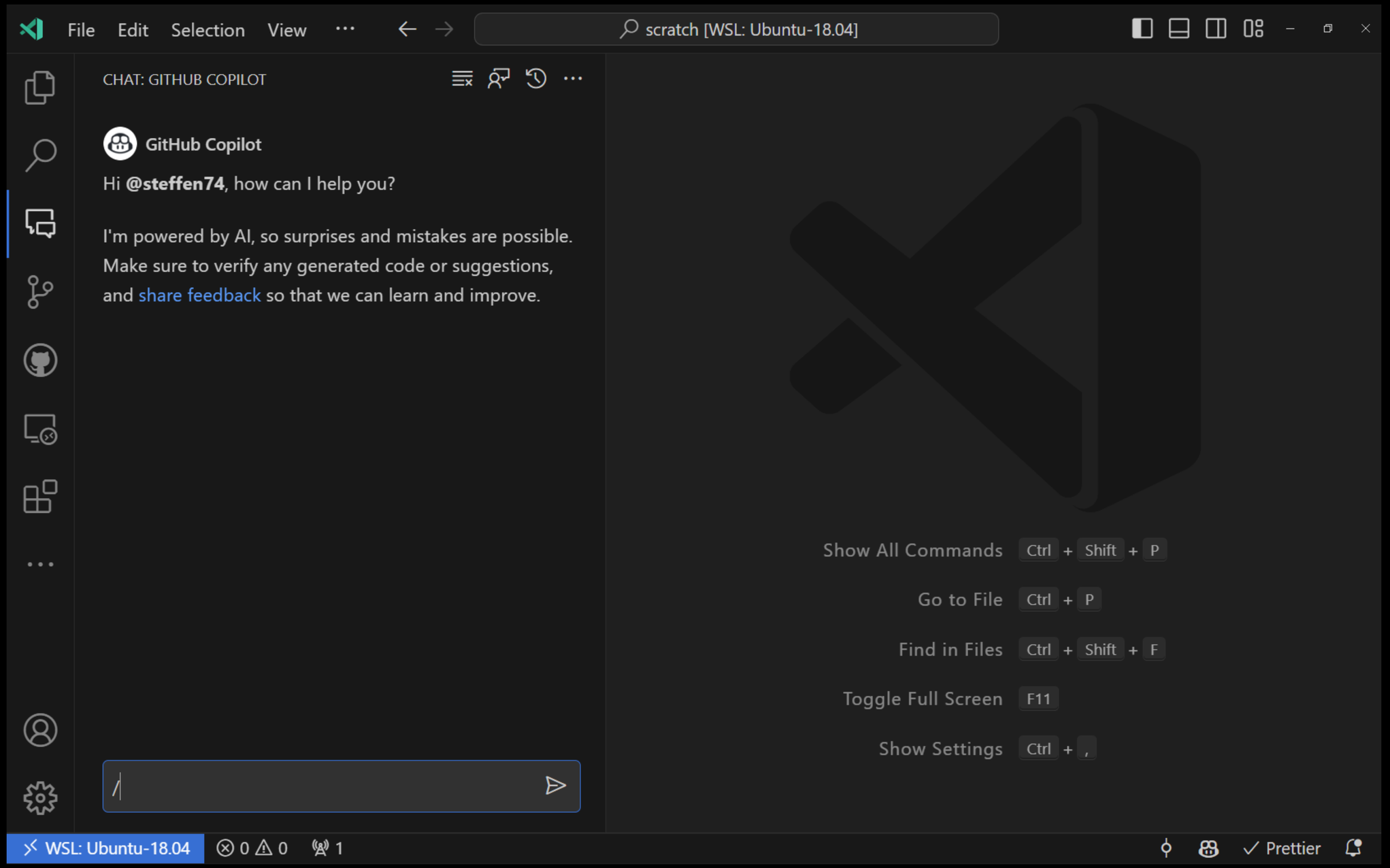
Message ChatGPT...

ChatGPT can make mistakes. Consider checking important information.

?


Spotify
Data Commons
indeed
instacart
Walmart
FiscalNote
replit
Bard
REDFIN
Adobe
WOLFRAM
Uber Eats
Khan Academy
ZipRecruiter
Tripadvisor
OpenTable
KAYAK

CHATBOTS ZUR PROGRAMMIERUNG


File
Edit
Selection
View
scratch [WSL: Ubuntu-18.04]
CHAT: GITHUB COPILOT
GitHub Copilot
Hi @steffen74, how can I help you?
I'm powered by AI, so surprises and mistakes are possible.
Make sure to verify any generated code or suggestions,
and share feedback so that we can learn and improve.
/
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Find in Files Ctrl + Shift + F
Toggle Full Screen F11
Show Settings Ctrl + ,
WSL: Ubuntu-18.04
0 0
1
Prettier

Merged **Adds New Tablegrid generic component** #809

ericstumper merged 14 commits into `staging` from `develop` 19 hours ago

review-agent-prime bot commented 20 hours ago

frontend-nx/apps/edu-hub/queries/generated /DegreeParticipantsWithDegreeEnrollments.ts

Consider adding comments to the newly added code to improve readability and maintainability. This will help other developers understand the purpose of the code and any important details they should be aware of.

Create Issue

```
// This is an example of how you might comment the new code
// URL to the file containing the user's attendance certificate (if he obtained one)
attendanceCertificateURL: string | null;
```

frontend-nx/apps/edu-hub/queries/user.ts

Consider optimizing the query by only fetching the necessary fields. This will reduce the amount of data transferred and improve the performance of the application.

Create Issue

```
// This is an example of how you might optimize the query
User(where: { status: { _eq: ACTIVE } }) {
  id
  firstName
  lastName
}
```

Consider sanitizing the input to the query to prevent SQL injection attacks. This can be done by using parameterized queries or prepared statements.

Create Issue

```
// This is an example of how you might sanitize the input
query UserForSelection1($searchValue: String!) {
  User(where: { firstName: { _ilike: $searchValue } }) {
    id
    firstName
    lastName
```

MULTIMODALE MODELLE

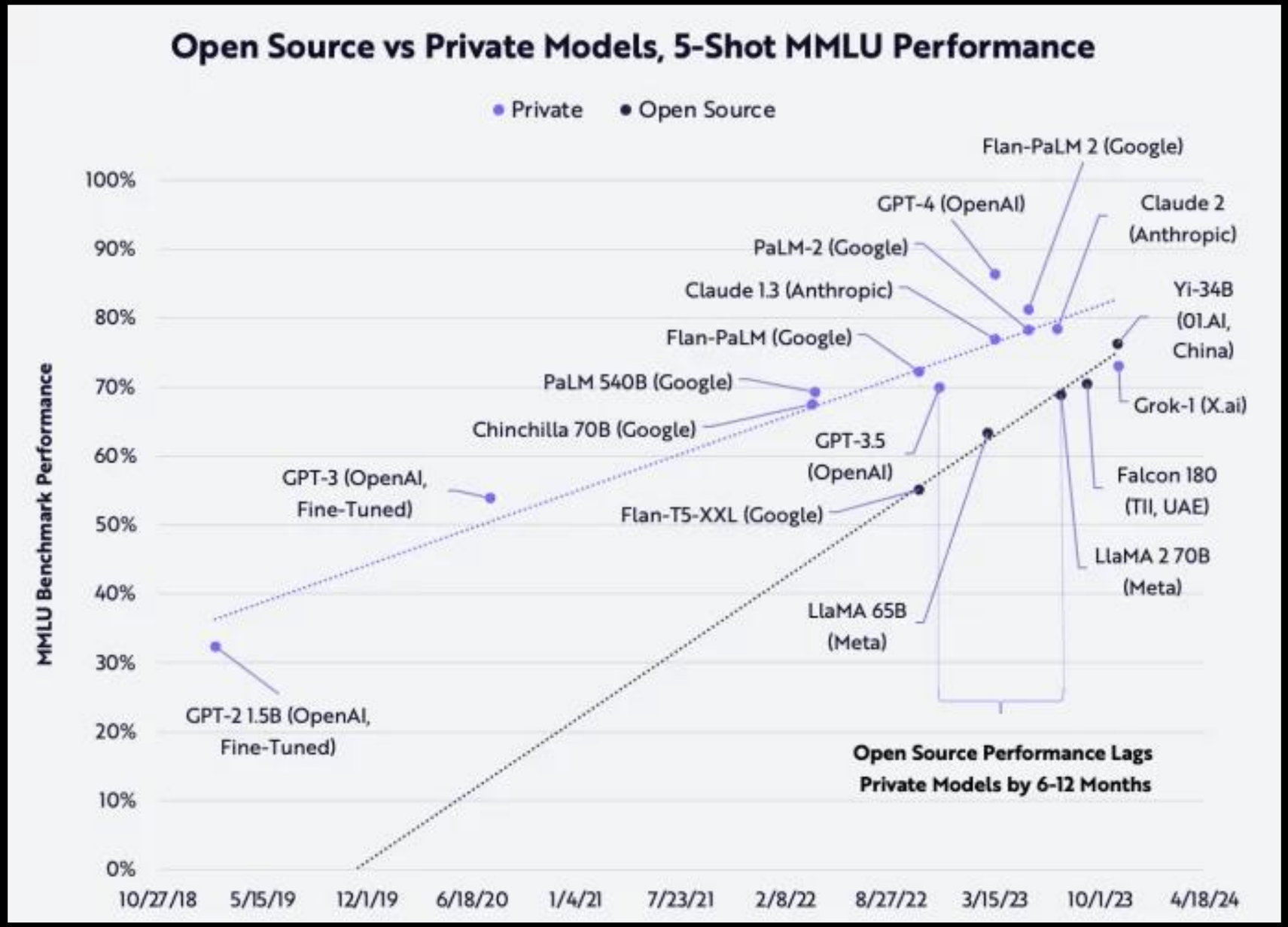


KREATIVITÄT

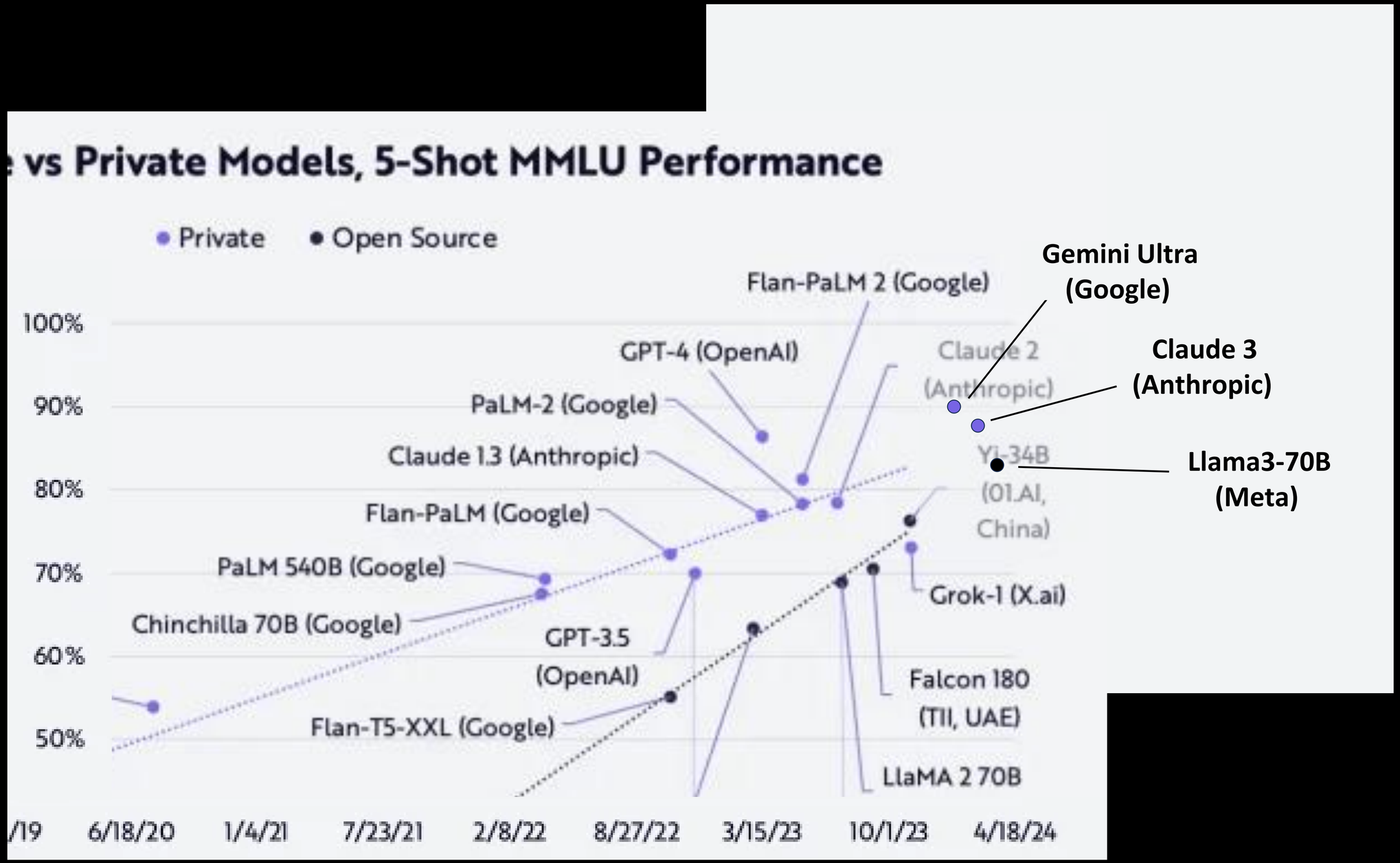
“A BOWL OF SOUP THAT LOOKS LIKE A MONSTER KNITTED OUT OF WOOL.”

“TEDDY BEARS MIXING SPARKLING CHEMICALS AS MAD SCIENTISTS IN A STEAMPUNK STYLE.”

OPEN SOURCE KI



ARK Invest. (2024). *BIG IDEAS 2024* [Annual Research Report]. ARK Investment Management LLC.

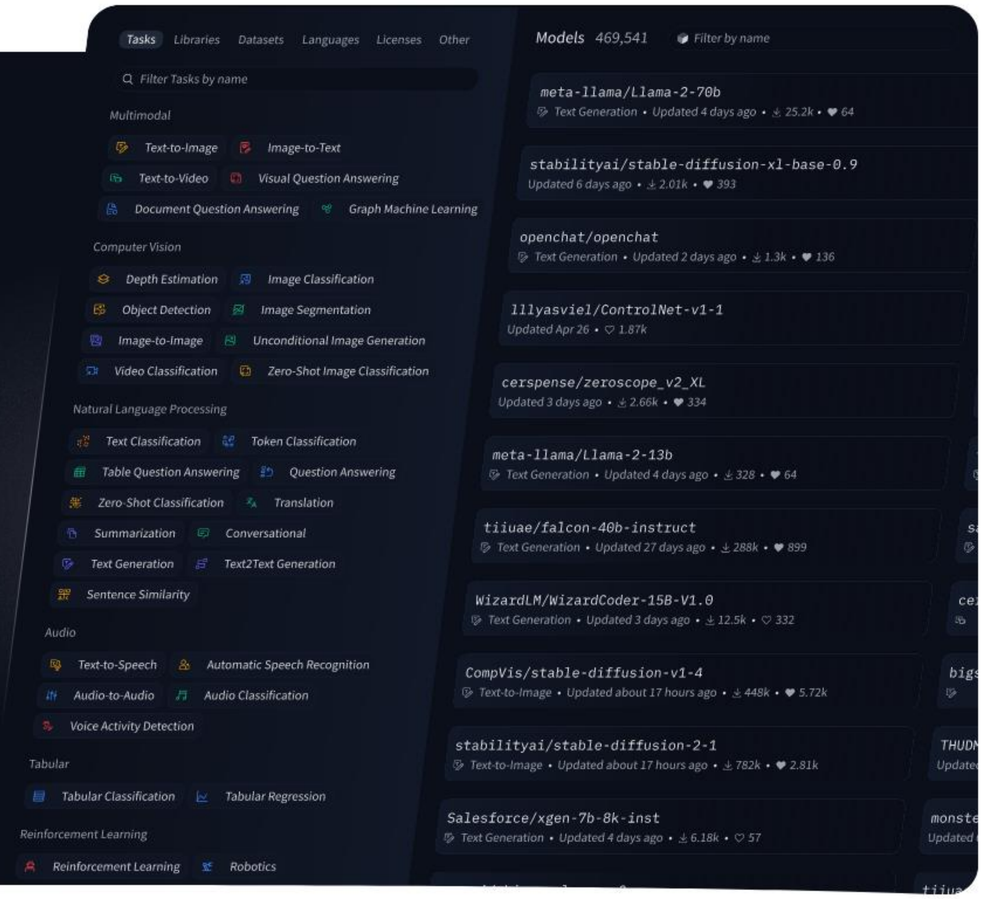

vs Private Models, 5-Shot MMLU Performance
Private
Open Source
100%
90%
80%
70%
60%
50%
Flan-PaLM 2 (Google)
GPT-4 (OpenAI)
Claude 2 (Anthropic)
PaLM-2 (Google)
Claude 1.3 (Anthropic)
Yi-34B (01.AI, China)
Flan-PaLM (Google)
PaLM 540B (Google)
Grok-1 (X.ai)
Chinchilla 70B (Google)
GPT-3.5 (OpenAI)
Falcon 180 (TII, UAE)
Flan-T5-XXL (Google)
LlaMA 2 70B
/19
6/18/20
1/4/21
7/23/21
2/8/22
8/27/22
3/15/23
10/1/23
4/18/24
Gemini Ultra (Google)
Claude 3 (Anthropic)
Llama3-70B (Meta)

Hugging Face | Search models, datasets, users... | Models | Datasets | Spaces | Posts | Docs | Solutions | Pricing | Log In | Sign Up

NEW Try Cohere Command R+ on HuggingChat

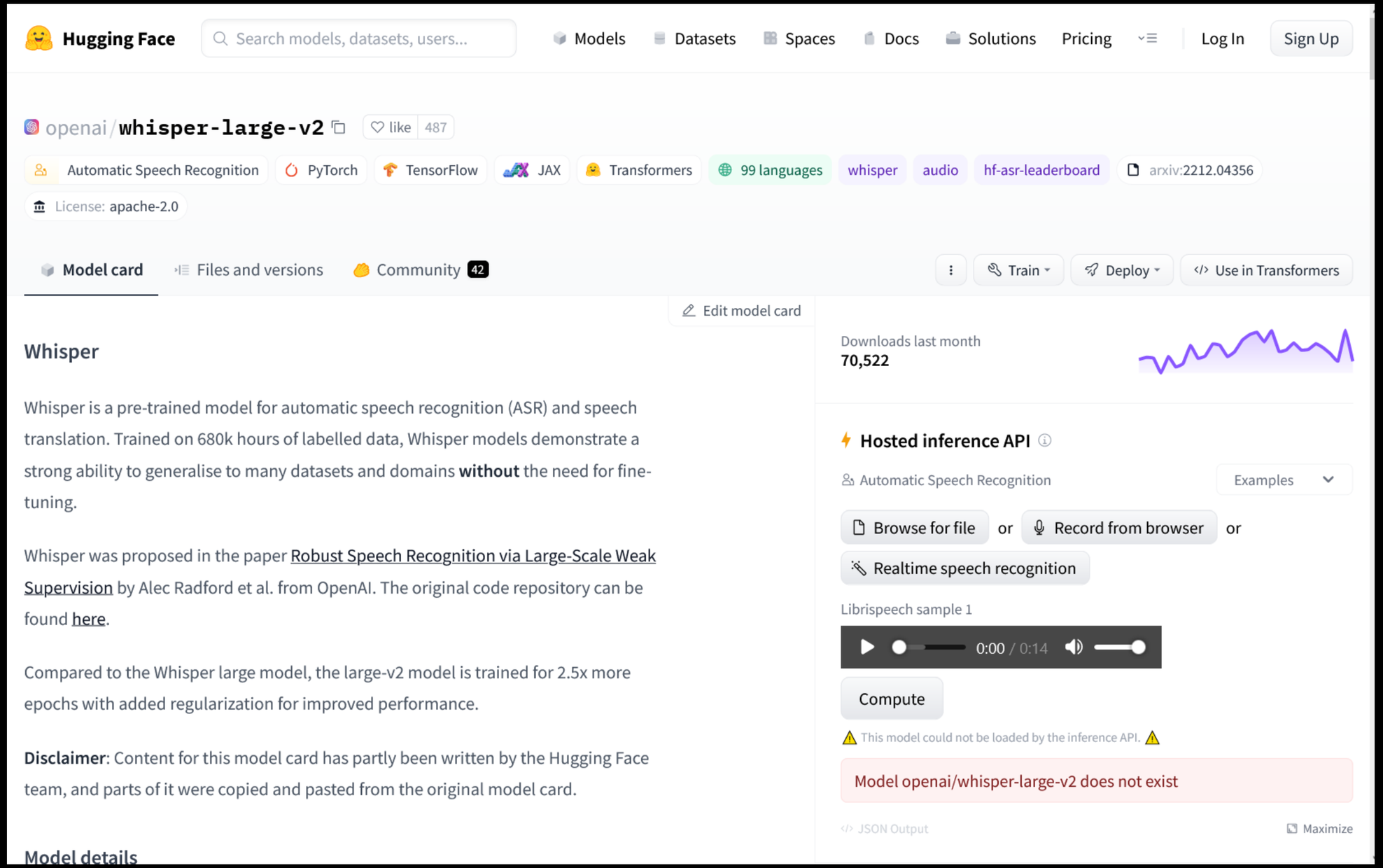
The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.



Trending on this week

Models Spaces Datasets


Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Docs
Solutions
Pricing
Log In
Sign Up
openai / whisper-large-v2
like 487
Automatic Speech Recognition
PyTorch
TensorFlow
JAX
Transformers
99 languages
whisper
audio
hf-asr-leaderboard
arxiv:2212.04356
License: apache-2.0
Model card
Files and versions
Community 42
Train
Deploy
Use in Transformers
Edit model card
Whisper
Whisper is a pre-trained model for automatic speech recognition (ASR) and speech translation. Trained on 680k hours of labelled data, Whisper models demonstrate a strong ability to generalise to many datasets and domains without the need for fine-tuning.
Whisper was proposed in the paper Robust Speech Recognition via Large-Scale Weak Supervision by Alec Radford et al. from OpenAI. The original code repository can be found here.
Compared to the Whisper large model, the large-v2 model is trained for 2.5x more epochs with added regularization for improved performance.
Disclaimer: Content for this model card has partly been written by the Hugging Face team, and parts of it were copied and pasted from the original model card.
Model details
Downloads last month
70,522
Hosted inference API
Automatic Speech Recognition
Examples
Browse for file
or
Record from browser
or
Realtime speech recognition
Librispeech sample 1
0:00 / 0:14
Compute
This model could not be loaded by the inference API.
Model openai/whisper-large-v2 does not exist
JSON Output
Maximize